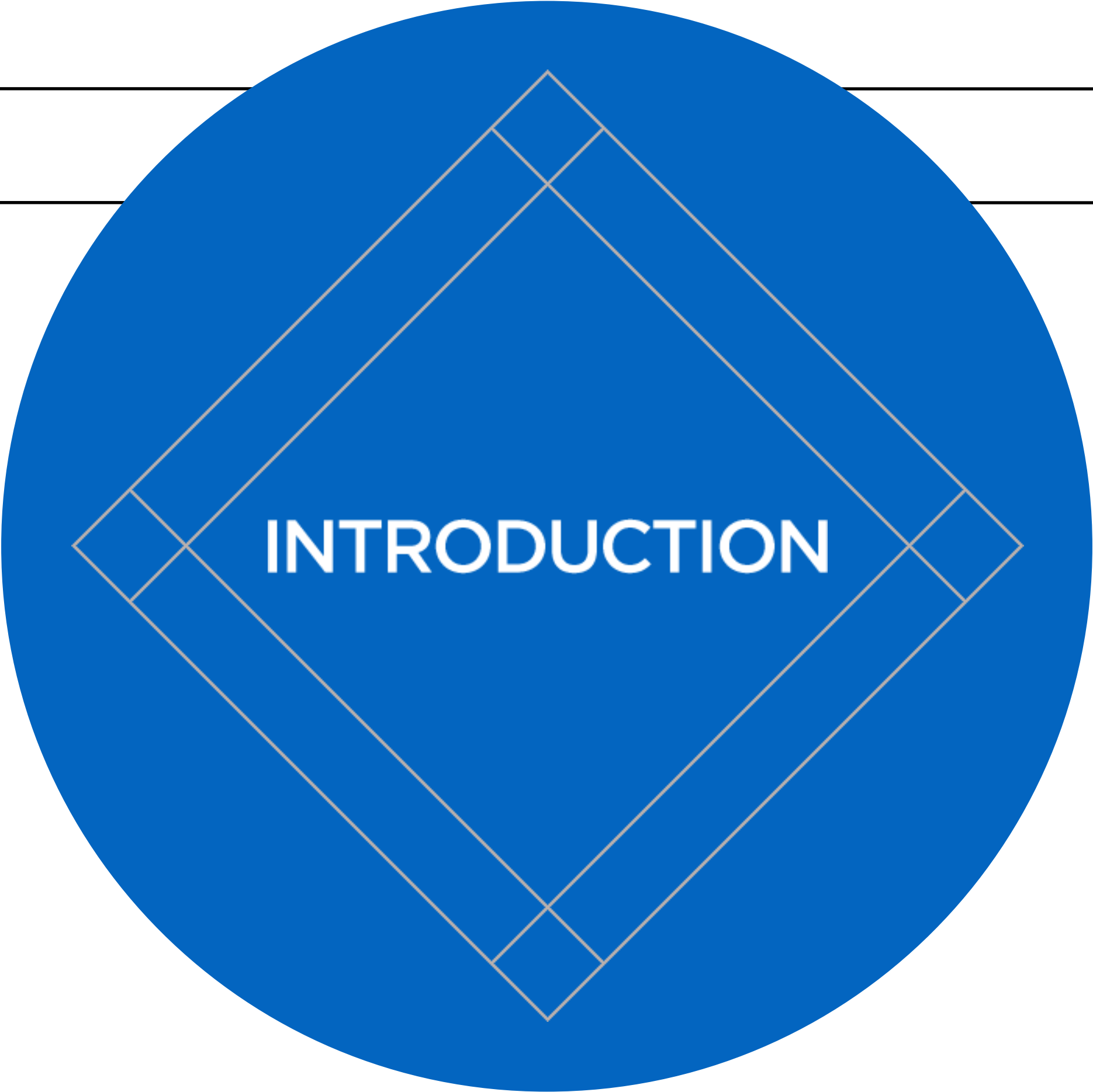


4

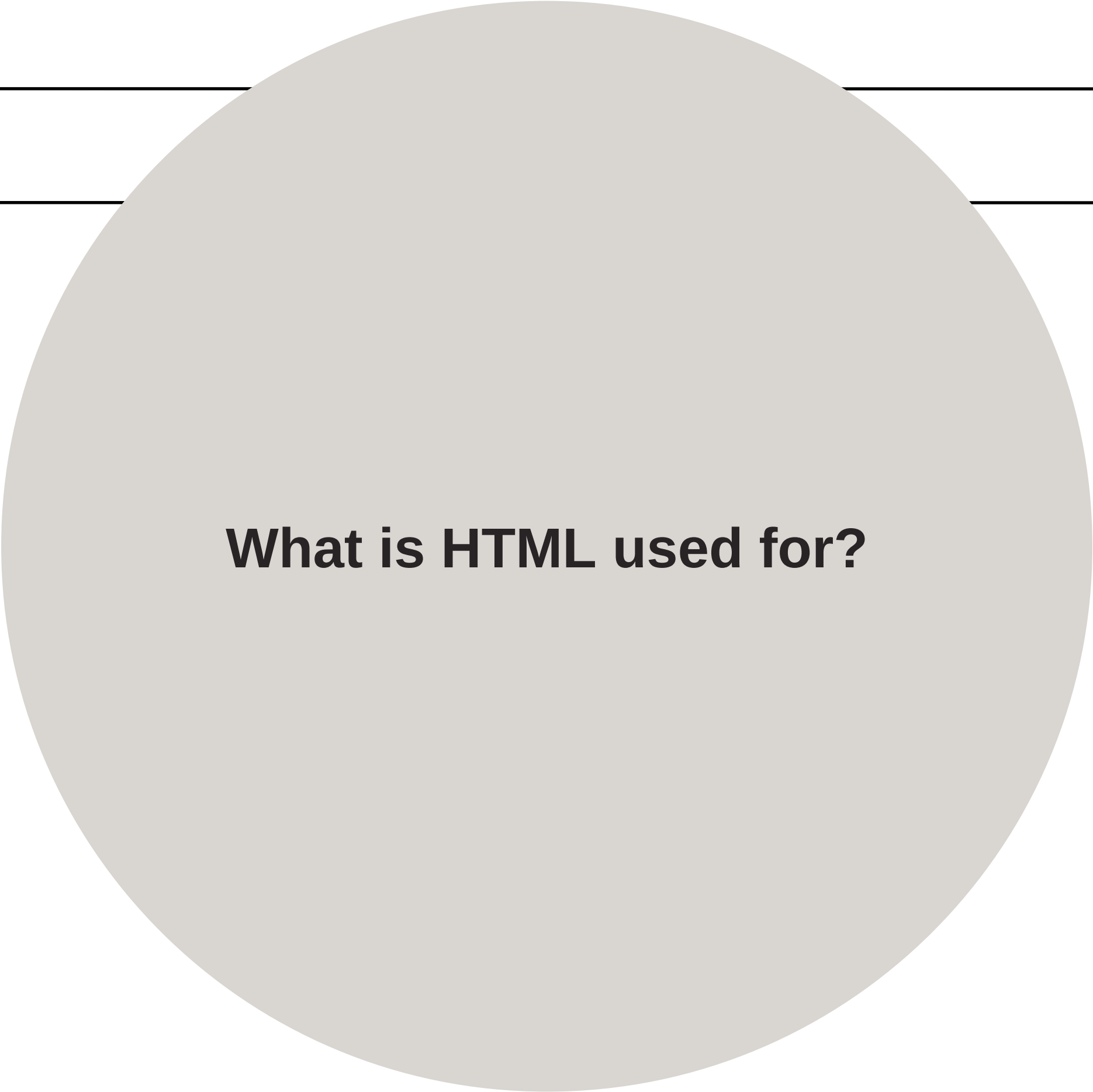
CREATING A SIMPLE PAGE (HTML OVERVIEW)

OVERVIEW

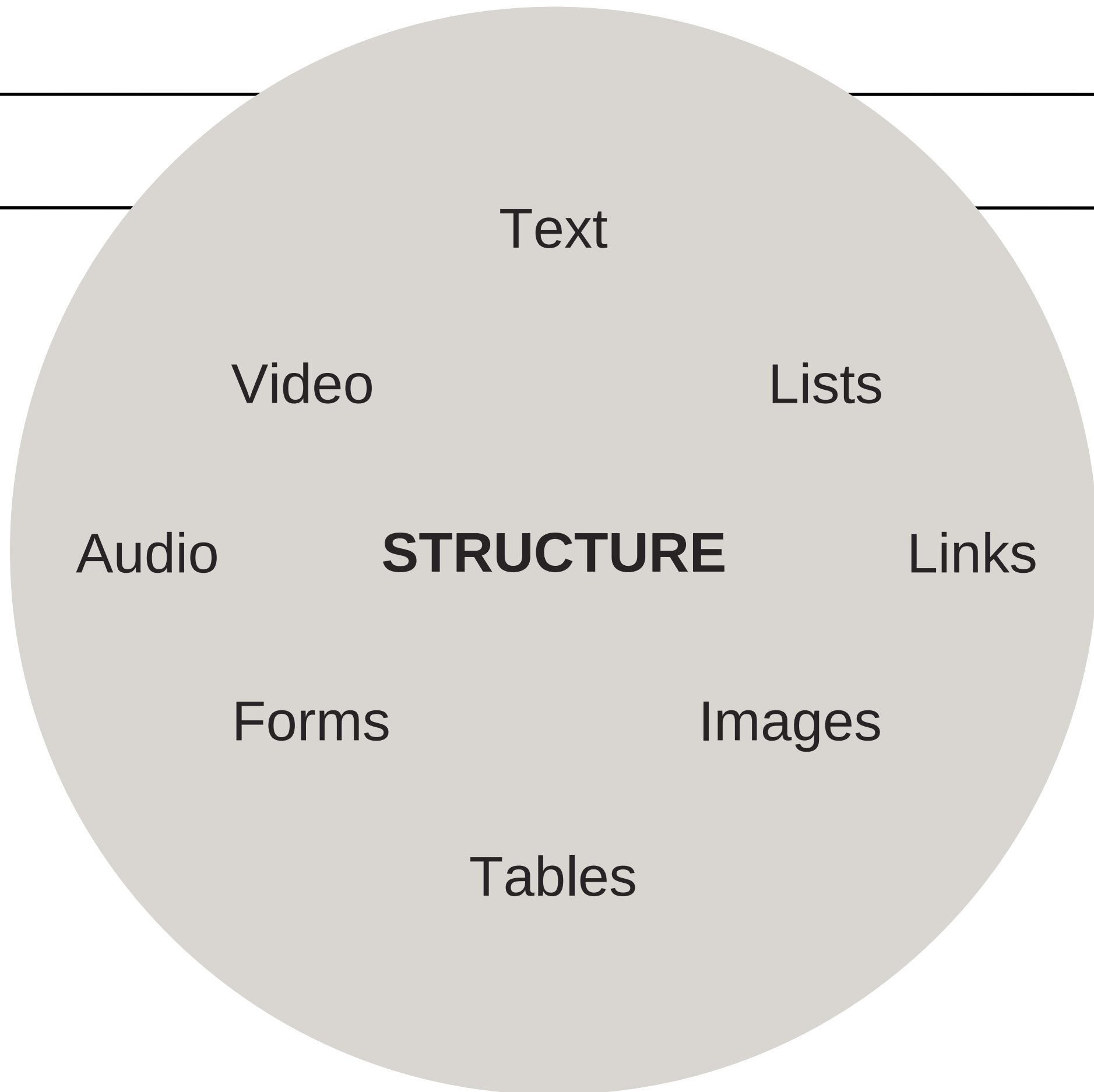
- **How markup works: Elements and attributes**
- **Minimal HTML document structure**
- **Identifying text elements: Block and inline**
 - Take space, margin, new line
- **Troubleshooting and validating HTML**
 - margin is according to need

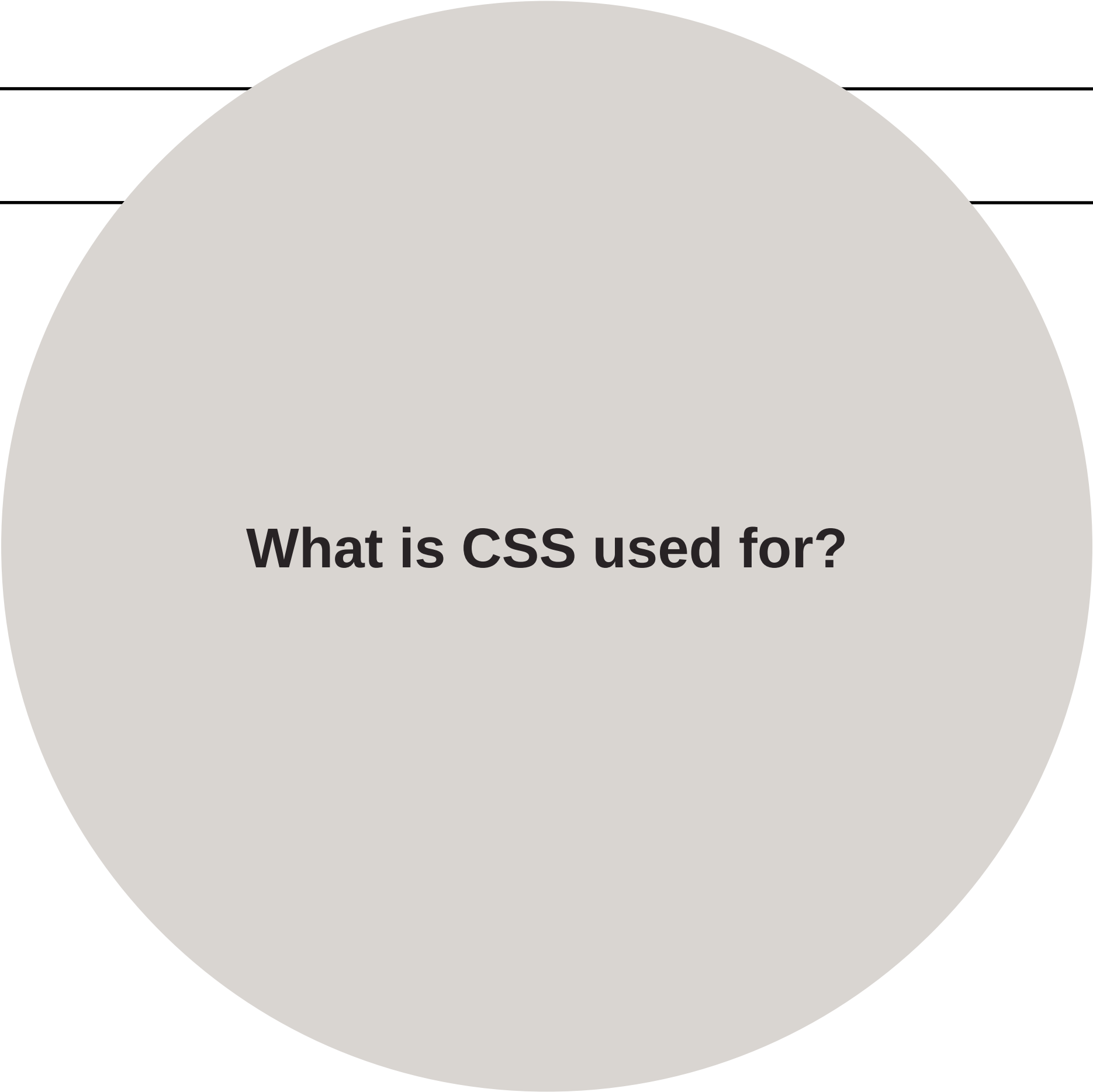


INTRODUCTION



What is HTML used for?



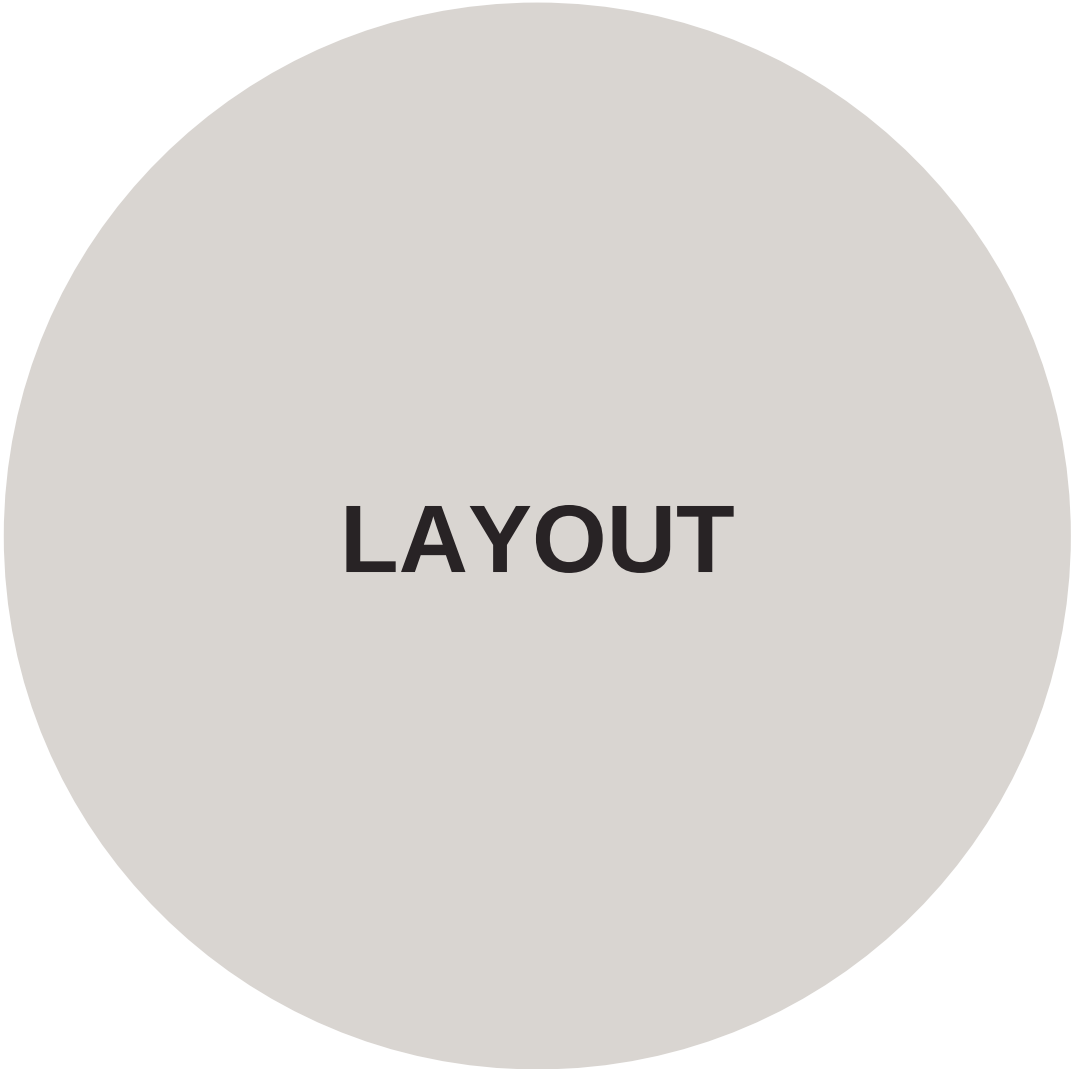


What is CSS used for?

OVERVIEW



PRESENTATION



LAYOUT

Content Without Markup

Without HTML markup to describe content structure, text runs together; line breaks are ignored:

Black Goose Bistro

The Restaurant

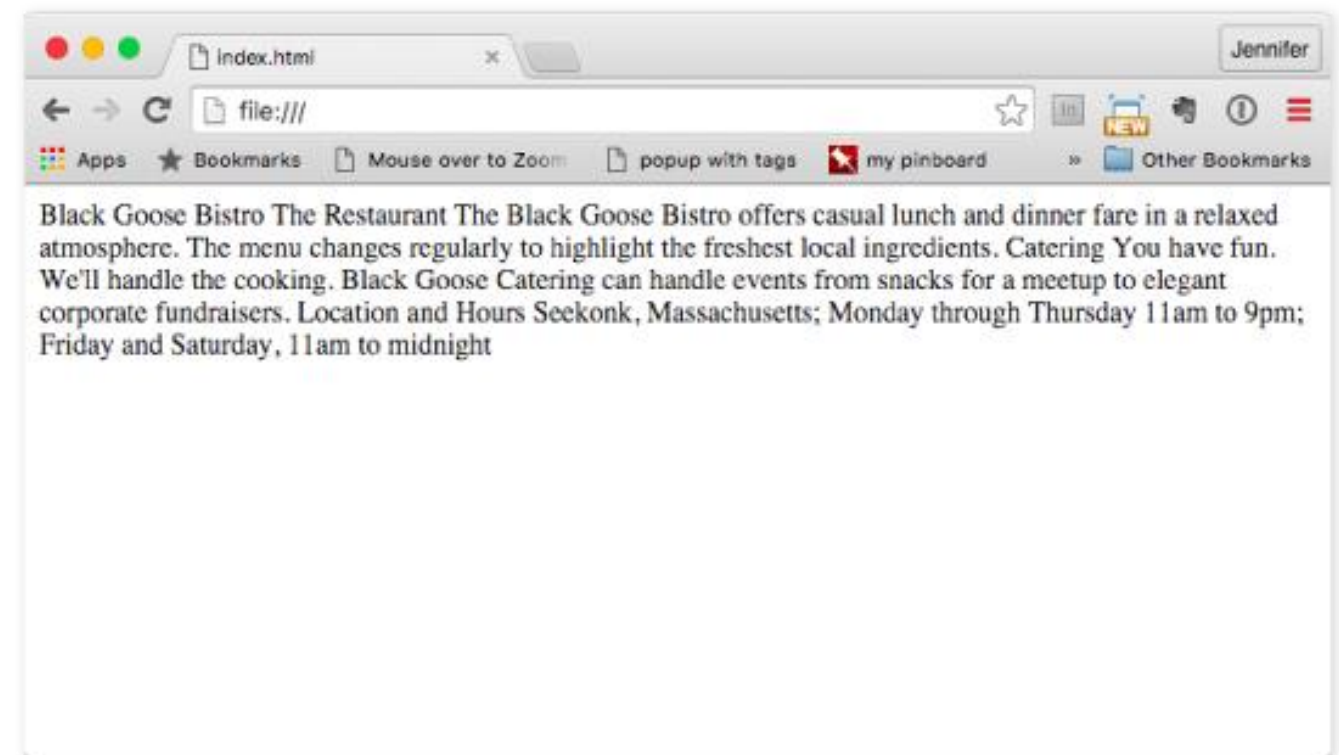
The Black Goose Bistro offers casual lunch and dinner fare in a relaxed atmosphere. The menu changes regularly to highlight the freshest local ingredients.

Catering

You have fun. We'll handle the cooking. Black Goose Catering can handle events from snacks for a meetup to elegant corporate fundraisers.

Location and Hours

Seekonk, Massachusetts;
Monday through Thursday 11am to 9pm; Friday and Saturday, 11am to midnight



What Browsers Ignore

- Multiple character spaces (white space)
- Line breaks (carriage returns)
- Tabs
- Unrecognized markup

Markup Basics

Text must be marked up meaningfully and accurately (**semantically**) with HTML tags:

- Browsers need markup to display content correctly.
- Markup makes content elements available to scripts and style rules.
- Markup aids search engines.

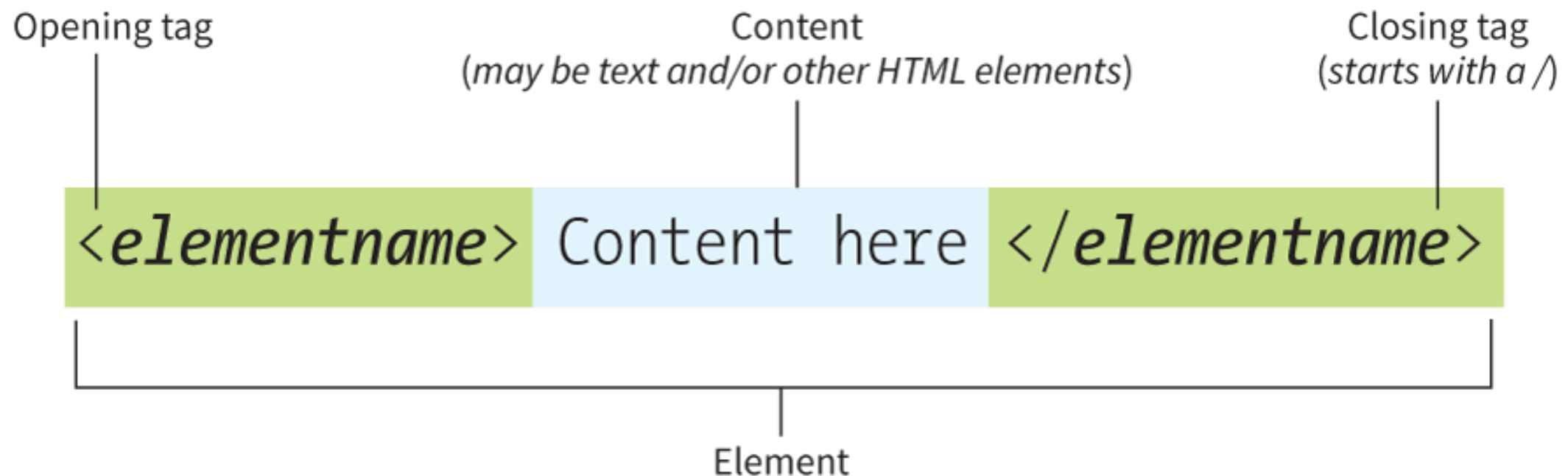
Anatomy of an HTML Element

tag

The element name in angle brackets

element

The content and its markup (start and end tags)



Example:

```
<h1>Black Goose Bistro</h1>
```

Some Elements Are Empty

Some elements have no content and provide a simple directive. They are called **empty elements**:

```
<element-name>
```

Example: The **br** element inserts a line break.

```
<p>1005 Gravenstein Highway North<br>Sebastopol, CA 95472</p>
```

Attributes

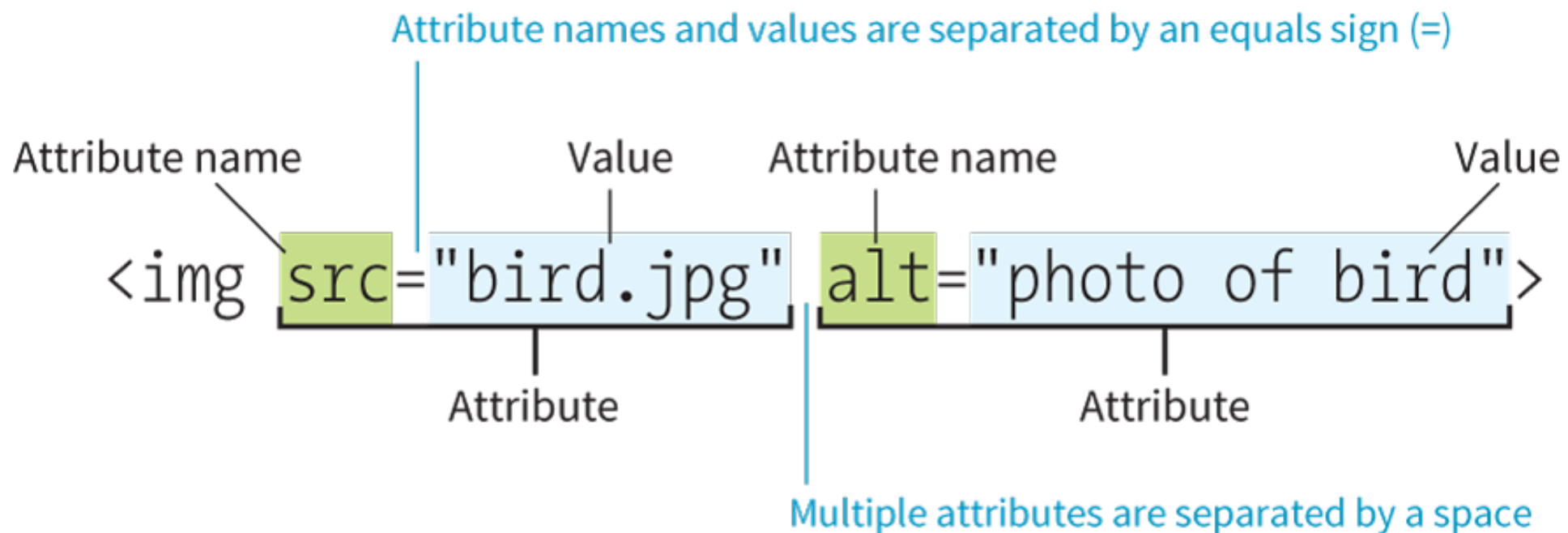
Attributes are instructions that clarify or modify an element. They appear in the opening tag after the element name:

```
<element attribute="value">Content</element>
```

```
<a href="http://oreilly.com">O'Reilly site</a>
```

Attributes (cont'd)

There can be more than one attribute in a tag:



They are separated by spaces and can go in any order.

Attributes (cont'd)

- Most attributes take values, which follow an = sign; some are single descriptive words.
- A value might be a number, word, string of text, URL, or measurement.
- Quotation marks aren't strictly required but are recommended for consistency.
- Single or double quotation marks are okay.
- Attribute names and values are defined in the HTML specification.
- Some attributes are required.

Nesting Elements

Putting elements inside other elements is called **nesting**. Make sure closing tags don't overlap:

```
<div>  
  <h1>Headline</h1>  
  <p>This is <em>emphasized</em> text.</p>  
</div>
```


HTML Comments

To leave a note in an HTML document, mark it up as a **comment**:

```
<!-- start global navigation -->
<ul>
...
</ul>
<!-- end global navigation
and begin the main header
for content page -->
```

Text between `<!--` and `-->` won't display in the browser. Keep in mind that they are still visible in the source.

Minimal Document Structure

It is recommended that HTML documents have the following minimal structure:

```
<!DOCTYPE html>
<html>

  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>

  <body>
    <!-- Page content goes here.-->
  </body>

</html>
```

Minimal Document Structure (cont'd)

The **DOCTYPE declaration** lets browsers know that the document is written according to the HTML5 specification:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Minimal Document Structure (cont'd)

The **html** element is the **root element** that contains all the elements in the document:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Minimal Document Structure (cont'd)

The **head** element contains elements that pertain to the document and are not rendered as content, such as **title**, **metadata**, **style sheets**, and **scripts**:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Minimal Document Structure (cont'd)

meta elements provide document **metadata**. In this case, it says that the document uses an expanded **character set** (UTF-8). It could also provide keywords, author information, publishing status, and more:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Minimal Document Structure (cont'd)

The **title** element is required. Titles display in the browser tab and bookmark lists, and are used by search engines:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```


Minimal Document Structure (cont'd.)

The **body** element contains the content of the document. The content is everything you want to show up in the browser window:

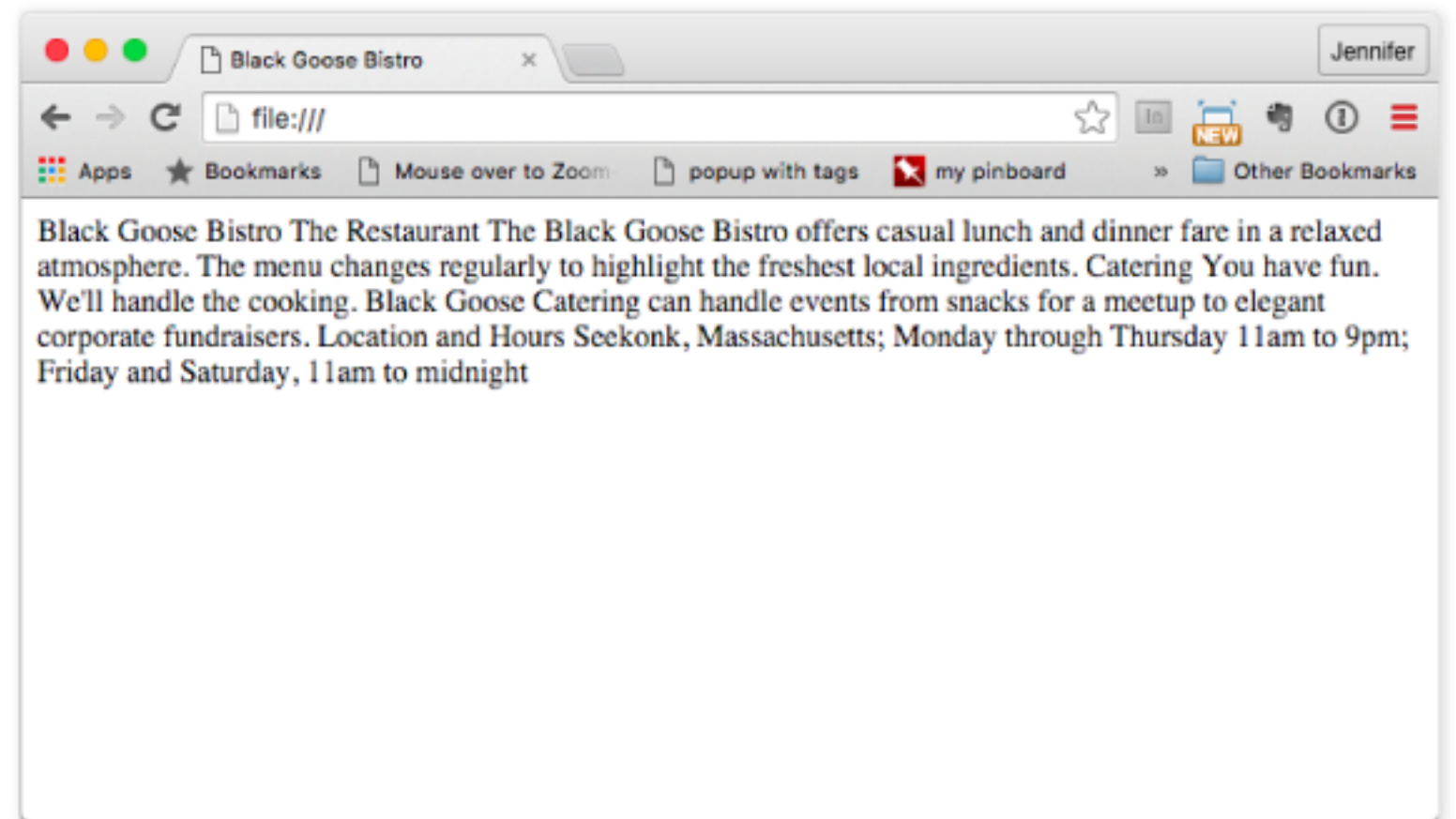
```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```


A Structured Document

With the document structure in place, there is a title in the browser tab, but the content is still unstructured.

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Black Goose Bistro</title>
</head>
<body>
Black Goose Bistro

The Restaurant
The Black Goose Bistro offers casual
lunch and dinner fare in a relaxed
atmosphere. The menu changes regularly
to highlight the freshest local
ingredients.
...
Location and Hours
Seekonk, Massachusetts;
Monday through Thursday 11am to 9pm;
Friday and Saturday, 11am to midnight
</body>
</html>
```



Marking Up Content

- The purpose of HTML is to add meaning and structure to the content. This is called **semantic markup**.
- It is **not** intended to describe how the content should look (its presentation).
- The way elements relate to one another forms an outline-like structure called the **DOM (Document Object Model)**.
- The DOM is the foundation on which you apply styles and scripts.

Basic Text Elements

With headings (<h1>) and paragraphs (<p>) identified, the browser can display the content appropriately:

...

```
<body>
```

```
<h1>Black Goose Bistro</h1>
```

```
<h2>The Restaurant</h2>
```

```
<p>The Black Goose Bistro  
offers casual lunch and  
dinner fare in a relaxed  
atmosphere. The menu changes  
regularly to highlight the  
freshest local  
ingredients.</p>
```

```
<!--more content-->
```

```
</body>
```

